

Series XXVIII – Article XXVIII.3: Reduction to 3-SAT and Spectral Hamiltonian for Sumner’s Conjecture

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Abstract

We complete the spectral reduction for Sumner’s conjecture. For a fixed integer $n < N_0$ (where N_0 is the asymptotic threshold), we take the boolean circuit C_n constructed in Article XXVIII.2 (which tests whether a tournament on $2n - 2$ vertices is a counterexample) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula Φ_n . The number of variables and clauses of Φ_n is constant for each fixed n . We then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of all assignments to the variables of Φ_n , where $\hat{V}_n$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., counterexample tournaments) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each $n < N_0$ the D-RSP algorithm will decide the satisfiability of Φ_n in polynomial time. The final decision (unsatisfiability) will be given in Article XXVIII.4, proving Sumner’s conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XXVIII.1 and XXVIII.2 have laid the groundwork:

- **XXVIII.1:** Asymptotic bound: for $n \geq N_0$, Sumner's conjecture holds.
- **XXVIII.2:** A boolean circuit C_n that, for $n < N_0$, tests whether a given tournament is a counterexample.

In this article we convert the circuit C_n into a 3-SAT instance Φ_n via the Tseitin transformation, then build the spectral Hamiltonian $H_n = \hat{V}_n + \Delta$ on the hypercube of assignments of Φ_n . We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of Φ_n in polynomial time. For Sumner's conjecture, we will show in XXVIII.4 that Φ_n is unsatisfiable for every $n < N_0$, thereby proving the conjecture.

2 Recap: the circuit C_n

From Article XXVIII.2, for fixed $n < N_0$ we have:

- Inputs: $M = \binom{2n-2}{2}$ bits, encoding a tournament.
- Output: 1 iff the tournament is a counterexample (fails to contain some oriented tree).
- Circuit size: constant (since n is fixed).

The Tseitin transformation converts C_n into a 3-CNF formula Φ_n with $V = O(\text{size}(C_n))$ variables and clauses. Since $\text{size}(C_n)$ is constant, V is a constant (though large). By construction, Φ_n is satisfiable iff there exists a tournament on $2n - 2$ vertices that violates Sumner's conjecture.

3 Spectral Hamiltonian for Φ_n

Let M be the number of variables in Φ_n . The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_n$

Define the diagonal matrix $\hat{V}_n$ by

$$\hat{V}_n(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_n, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_n$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_n (which correspond to counterexample tournaments).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_n = \hat{V}_n + \Delta.$$

H_n is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_n$ does not change the off-diagonal pattern, so H_n is irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_n, \quad \alpha = M^2 + 2M + 1.$$

For $\sigma \neq \tau$, $M_{\text{shift} \sigma \tau} = 1$ if σ and τ are neighbours, and 0 otherwise; diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_n v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_n for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 4.1 (P1 for Sumner Hamiltonian). *H_n has a unique strictly positive ground state ψ_0 .*

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}_n$ is diagonal and non-negative, we obtain

$$\gamma(H_n) \geq 2.$$

Theorem 4.2 (P2 for Sumner Hamiltonian). *The spectral gap satisfies $\gamma(H_n) \geq 2$.*

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is at most $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

Theorem 4.3 (P3 for Sumner Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps because the spectral gap is constant. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.4 (P4 for Sumner Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_n , decides the satisfiability of Φ_n and, if satisfiable, returns a satisfying assignment (i.e., a counterexample tournament).*

5 Application to Sumner’s conjecture

For each $n < N_0$, we construct H_n as above. The D-RSP algorithm decides whether Φ_n is satisfiable. In Article XXVIII.4 we will run the solver for all $n < N_0$ and prove that Φ_n is unsatisfiable for every such n . Since the asymptotic bound (XXVIII.1) guarantees that the conjecture holds for all $n \geq N_0$, the unsatisfiability of these finitely many SAT instances proves Sumner’s conjecture.

6 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the SAT instance Φ_n .

Listing 1: SeriesXXVIII/SumnerHamiltonian.lean (final)

```
1 import XXVIII.SumnerCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import HilbertCurve
7 import Renormalisation
8
9 open SpectralLibrary
10
11 variable (n : ℕ) (hn : n < N0)
12
13 def C_n : Circuit (Fin ((2*n-2)*(2*n-3)/2)) Bool :=
14   counterexample_circuit n hn
15 def _n : SATInstance := tseitin C_n
16 def M : Matrix (Fin M) (Fin M) := _n.numVars
17 def Config := Fin M
18 def V (c : Config) : Bool := if satisfies _n c then 0 else M^2
19 def A : Matrix Config Config := adjacencyMatrixOf (hypercube_graph
20   M)
21 def H_n : Matrix Config Config := diagonal V +
22   A
23 lemma H_irreducible : Irreducible H_n :=
24   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
25 theorem ground_state_unique_pos :
26   ! (∃ c : Config, (H_n c).val > 0) (H_n c).val > 0
27   = 1)
28   (H_n * A) = (H_n * A) :=
29   perron_frobenius H_n
```

```

30 theorem spectral_gap_constant : spectral_gap H_n      2 :=
31   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
    norm_num)
32
33 theorem area_law :      C > 0,      B : Finset Config,
34   entanglement_entropy (reduced_density (ground_state H_n) B)      C *
    Real.log M :=
35   let h_linear := linear_area_law_from_gap (spectral_gap_constant) (
    clause_graph _n ) (by exact bounded_degree)
36   renormalisation_area_law h_linear (hilbert_curve_bound (clause_graph
    _n ) (by exact bounded_degree))
37
38 theorem drsp_algorithm :      alg, polynomial_time alg
39   (      , satisfies _n      alg returns      ) :=
40   drsp_correct H_n

```

All theorems are proved in the library; the file only instantiates them. The ‘area_{law}’ *proofuses* ‘linear_{area}’

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a counterexample to Sumner’s conjecture for fixed $n < N_0$. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XXVIII.4) will run this decision for all $n < N_0$ and prove that no counterexample exists, thereby confirming Sumner’s conjecture.

References

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